The Grammar as the $\Sigma \to 1:1$ Limit of the Belnap Multilattice SIC-POVM

The structural SIC-POVM is already solved — the open part is Hilbert-space embedding

Belnap Multilattice SIC-POVM

$\Sigma = n : m$ (many-to-many)

- 4-valued Belnap logic (T, B, F, N)
- All 16 Belnap states
- meet$(B, x) = x$, join$(B, x) = B$
- bnot$(B) = B$ unconditional
- SIC-POVM axioms hold

Distance to Grammar: $d = 2.0$
Only Σ differs: $n : m$ *vs* $1 : 1$

Fiducial: $B_{\text{Belnap}} = XZ$

$\Sigma : \; n : m \to$

Self-reference

Imscribing Grammar SIC-POVM

$\Sigma =$ ϙ (1:1)

- 12 primitives as measurement operators
- 6 Frobenius-dual pairs
- $\mu \circ \delta =$ id for every pair
- $\odot$ criticality gate open (Gate 1)
- K $\leq \iota$ gate open (Gate 2)
- Crystal of Types $= 17.28$M outcomes

Tier: O_∞ **C-score:** 1.0

Fiducial: $B_{\text{Grammar}} = XZ$

Lean 4 Verification: 22 theorems, 0 sorries

`BelnapNFiducial.lean` — fiducial axioms proven `SIC_Multilattice_Proof.lean` — SIC-POVM multilattice theorem
`ZaunerEmbeddingEquivalence.lean` — Zauner conjecture structural embedding